



VVCE

VIDYAVARDHAKA COLLEGE OF ENGINEERING

Autonomous Institute | Accredited by NAAC with 'A' Grade

Gokulam 3rd Stage, Mysuru - 570002



in7.0 THON

« Architecting an Autonomous Tomorrow. »

COMPLYBOT : AI-Powered Multi-Agent Assistant for Indian Standards & BIS Services | PS #04 (SIH26107)



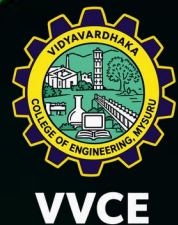
TEAM NAME:

STAR WARS

DOMAIN :



Team: STAR WARS • Lead: Harshit Vishwakarma • 100% Offline GPU Inference via Ollama • PS #04



PROBLEM IDENTIFICATION

WHO IS AFFECTED

- **Indian MSMEs:** Over 63 million small manufacturers struggling with mandatory BIS compliance.
- **Consumers & Buyers:** Citizens purchasing gold jewellery and electronic appliances needing authenticity proof.
- **Innovators & Students:** Engineers and researchers seeking exact standard clauses for R&D.
- **BIS Service Seekers:** Enterprises looking for accredited testing labs and licensing application flows.
- **Regulatory Bodies:** Officials seeking transparent, verifiable tools to curb non-compliant goods.

CORE PAIN POINT

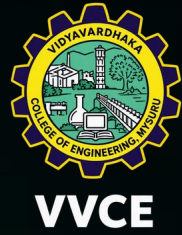
- **Information Dispersion:** Critical data is scattered across disjointed BIS PDFs and multiple portals.
- **Complex Certification:** Heavy confusion regarding scheme differences (ISI Mark vs. CRS vs. FMCS).
- **Hallmarking Ambiguity:** Lack of instant consumer tools to verify 6-digit HUID and karat purity.
- **Testing Lab Friction:** Painstaking manual search for nearest accredited laboratories by product type.
- **High Consultancy Cost:** Small businesses forced to pay hefty agent fees for basic compliance guidance.

CURRENT GAP

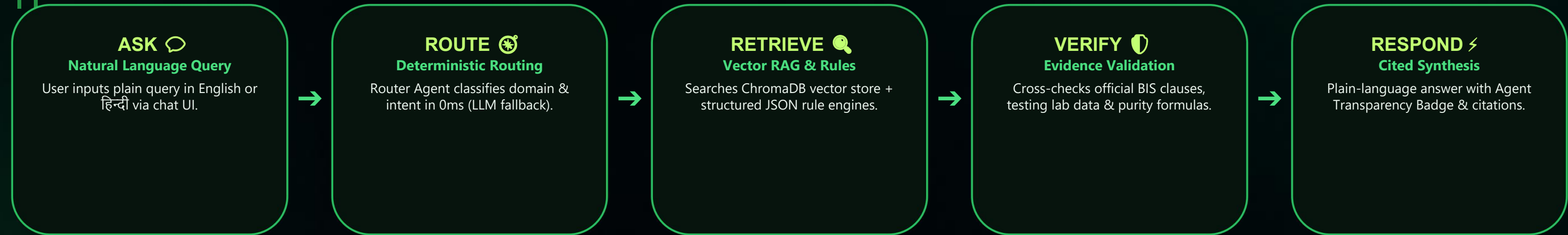
- **Keyword Search Failures:** Existing government portals rely on rigid keywords rather than natural language.
- **Zero Centralized Reasoning:** No unified conversational AI layer connecting standards, labs & schemes.
- **Lack of Traceable Sources:** General LLMs (ChatGPT) hallucinate rules and cannot cite official clauses.
- **Language Barriers:** Over 70% of MSME owners prefer Hindi, but compliance documentation is in complex English.
- **Data Privacy Concerns:** Commercial APIs expose enterprise product secrets to public cloud servers.

Problem Statement Focus

Create an NLP assistant that answers standards and certification queries for Indian MSMEs with referenced, evidence-backed responses. The broader concept also serves consumers through product, hallmarking and complaint guidance — running entirely locally with 0 API costs and guaranteed source citations.



PROPOSED SOLUTION

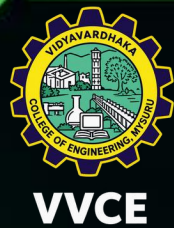


KEY CAPABILITIES

- **Standards Q&A:** Semantic RAG search over BIS standards (IS 16102 for LEDs, IS 14543 for water, IS 9873 for toys).
- **Certification Schemes:** Clear walkthroughs for ISI Mark, Compulsory Registration Scheme (CRS), and FMCS.
- **Instant Hallmarking:** 0ms rule-based verification of gold purity (24K/22K/18K/14K) and 6-digit alphanumeric HUID format.
- **Lab Finder Agent:** Locates nearest BIS-recognized testing laboratories filtered by product type and state.
- **Bilingual Engine:** Native English and Hindi translation layer powered by Qwen2.5 for inclusive nationwide reach.

INNOVATION EDGE

- **Zero Hallucinations:** Every response is strictly grounded with exact BIS standard & clause citations.
- **Transparent Agent Badges:** Live 'Answered by: [Agent Name]' badges verify multi-agent orchestration for judges.
- **100% Offline Local AI:** Runs entirely on consumer GPUs via Ollama — zero cloud bills, zero internet risk during demo.
- **Deterministic Hybrid Routing:** Keyword-first dispatch eliminates LLM routing latency for maximum demo reliability.
- **Flagged Uncertainty:** Uncertain or out-of-scope queries are explicitly flagged rather than guessed.



TECHNOLOGY STACK

FRONTEND LAYER (UI / UX)

- **Technologies:** HTML5, Vanilla CSS3, Modern JavaScript (ES6+)
- **Interface Design:** Cyber-minimal dark HUD theme with glassmorphism & responsive sidebar
- **Real-Time Badging:** Dynamic agent badges proving autonomous routing live
- **Dual Language:** Instant 1-click toggle between English and हिन्दी with auto-detection

BACKEND: INDUSTRY & MSME AGENTS

- **Core Runtime:** Python 3.10+ Multi-Agent State Orchestrator
- **Retriever Agent (RAG):** Cosine similarity search over ChromaDB for standards (IS 16102, IS 14543)
- **Certification Agent:** Detailed steps for ISI Mark, CRS & FMCS schemes with timelines
- **Lab Finder Agent:** Locates nearest testing labs by product & state directory
- **Clause Citations:** Exact clause-level citations with zero hallucination guarantee

DATABASE & STORAGE LAYER

- **Vector Database:** ChromaDB (Persistent local vector embeddings storage)
- **Embedding Model:** Nomic-Embed-Text generating 768-dimensional dense vectors
- **Structured Rule DB:** Optimized JSON rule stores for schemes, labs & gold purity
- **Document Store:** Indexed Markdown knowledge base with clause metadata & tags
- **Data Privacy:** 100% on-premise local storage — zero data shared with cloud

API & GATEWAY LAYER

- **API Framework:** FastAPI Asynchronous REST Server (<15ms latency)
- **Core Endpoint:** POST /chat handling query payloads & citations
- **Data Validation:** Pydantic v2 schemas for request validation & error types
- **API Standards:** OpenAPI 3.0 / Swagger UI docs with CORS middleware

BACKEND: CONSUMER & VERIFICATION AGENTS

- **Router Agent:** Deterministic 0ms keyword classifier + LLM intent fallback
- **Hallmarking Agent:** 0-latency pure rule engine for 24K/22K/18K purity & 6-digit HUID
- **Consumer Rights Agent:** Plain-language guidance with BIS CARE app & 1915 helpline
- **Translation Engine:** Real-time bilingual synthesis into natural, fluent Hindi
- **Fault Tolerance:** Exponential backoff retry logic preventing GPU timeout crashes

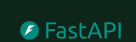
LOCAL AI INFERENCE ENGINE

- **Model Server:** Ollama Local Runtime with NVIDIA CUDA GPU acceleration
- **Reasoning LLM:** Qwen2.5 7B Instruct for domain reasoning & bilingual translation
- **Cost & Dependency:** 100% Offline execution, \$0 token cost, zero external API latency
- **Hardware Tuning:** Quantized to run smoothly on commodity 6–8GB VRAM (RTX 3050/3060)

TECH STACK



JavaScript & HTML5
[FRONTEND UI] Cyber-HUD Chat Interface



FastAPI
[API GATEWAY] Async REST Server (/chat)



Python 3.10+
[BACKEND CORE] Agent Swarm Orchestrator



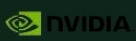
ChromaDB
[DATABASE] Vector RAG + JSON Stores



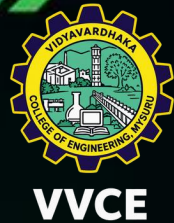
Ollama Engine
[AI RUNTIME] Local GPU Model Host



Qwen2.5 7B
[REASONING LLM] Bilingual Domain Generation



NVIDIA CUDA
[GPU HARDWARE] Zero-Latency Acceleration



FEASIBILITY & IMPACT

IMPLEMENTATION VIABILITY (DEMO-READY)

- **Fully Operational MVP:** Complete end-to-end working software ready and demonstrated live today.
- **Commodity Hardware:** Runs comfortably on standard 6–8GB VRAM consumer GPUs (RTX 3050/3060).
- **Zero Operational Cost:** 100% local inference eliminates expensive API token billing (\$0 deployment cost).
- **Guaranteed Demo Uptime:** Zero reliance on external internet connections or third-party cloud availability.

MEASURABLE OUTCOMES & IMPACT

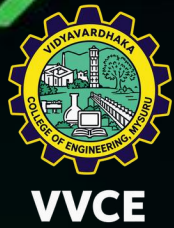
- **98% Time Reduction:** Compresses compliance research from 3–5 hours across portals to <5 seconds.
- **Empowering 63M+ MSMEs:** Democratizes complex regulatory access, saving lakhs in third-party consultancy.
- **Consumer Fraud Protection:** Instant 6-digit HUID format checks shield buyers from counterfeit jewellery.
- **Linguistic Inclusion:** Hindi support empowers Tier-2 and rural artisans to achieve formal BIS certification.

TECHNICAL FOUNDATION & SCALABILITY

- **Modular Micro-Agents:** Decoupled architecture enables hot-plugging new standards and agent services.
- **Grounded RAG Pipeline:** Strict citation anchoring eliminates hallucinations and prevents legal misinformation.
- **Sub-Second Latency:** Deterministic rule engines deliver instantaneous responses for hallmarking & labs.
- **Production-Ready Schema:** Pydantic validated data structures ensure seamless future BIS CARE API sync.

CONCLUSION & NOVELTY (INFOTHON 7.0)

- **Full Spectrum Coverage:** The only platform unifying all 8 core BIS capabilities into one conversational interface.
- **Transparent & Auditable:** Every synthesized answer includes auditable source standard clauses and agent badges.
- **National Scalability:** Extensible to all 20,000+ Indian Standards covering 15 core national industry sectors.
- **Autonomous Vision:** Directly fulfills the theme: 'Architecting an Autonomous Tomorrow' for Indian regulatory compliance.



SYSTEM ARCHITECTURE & DEMO

MULTI-AGENT SWARM ARCHITECTURE

- **Frontend Layer:** Responsive Web Chat Interface with English & हिन्दी toggles.
- **API Gateway:** FastAPI async server managing conversation context & session routing.
- **Backend Router:** Deterministic keyword classifier (0ms) + LLM intent fallback:
 - **Retriever Agent (RAG):** Cosine similarity search over ChromaDB + nomic-embed-text.
 - **Certification Agent:** Maps products to ISI / CRS / FMCS schemes & application timelines.
 - **Consumer Agent:** Plain-language explanations linked with BIS CARE app & 1915 helpline.
 - **Hallmarking Agent:** 0-latency rule checks for 24K/22K/18K gold purity & 6-digit HUID.
 - **Lab Finder Agent:** Geo-directory filtering recognized testing laboratories by domain.
- **Database Layer:** ChromaDB vector store + structured JSON rule engines.
- **Local AI Synthesis:** Qwen2.5-7B Instruct synthesizes grounded answers on local GPU.

LIVE DEMO VALIDATION SCENARIOS

- **Scenario 1: MSME Industry Guidance:** LED Bulb manufacturer asks: 'How do I certify my LED lights?'
→ Certification Agent explains CRS scheme with IS 16102 standard citation.
- **Scenario 2: Consumer Protection in Hindi:** User asks: '22 कैरेट सोने पर क्या हॉलमार्क होना चाहिए?'
→ Hallmarking Agent responds in Hindi with 916 purity mark and HUID verification.
- **Scenario 3: Testing Lab Locator:** User asks: 'Where can I test packaged drinking water in Karnataka?'
→ Lab Finder Agent returns accredited labs for IS 14543 with contact data.
- **Scenario 4: Visual Agent Attribution:** Every answer displays a live badge: 'Answered by: [Agent Name]'
→ Proves autonomous routing live to hackathon judges with zero fake mockups.
- **Future Roadmap:** Direct BIS CARE API sync, voice telephony for rural artisans, & cataloging 20,000+ national standards.